

Conceptual Structural Design of an Aircraft Wing Box

Robert Anadu

January 21, 2026

Abstract

This report presents a conceptual structural design study of an aircraft wing box intended for portfolio demonstration purposes. The work focuses on realistic structural layout, internal load paths, and manufacturability considerations representative of early-stage aerospace design. A parametric CAD model is developed to support the structural reasoning, while assumptions and limitations are stated explicitly to reflect professional engineering practice.

Contents

1	Introduction	4
2	Design Context and Assumptions	4
2.1	Design Context	4
2.2	Reference Aircraft Characteristics	5
2.3	Structural Design Philosophy	5
2.4	Material Assumptions	5
2.5	Geometric and Modeling Assumptions	6
2.6	Scope and Limitations	6
3	Flight Loads and Design Cases	6
3.1	Purpose of Load Definition	6
3.2	Load Factor Concept	6
3.3	Representative Design Load Factors	7
3.4	V–n Diagram Interpretation	7
3.5	Critical Structural Design Cases	7
3.6	Implications for Wing Box Design	8
3.7	Limitations of Load Definition	8
4	Wing Box Structural Design	8
4.1	Role of the Wing Box	8
4.2	Structural Concept	8
4.3	Geometric Configuration	9
4.4	Planform and Cross-Section Views	9
4.5	Internal Structure and Section Analysis	10
4.6	Load Path Discussion	11
4.7	Structural Efficiency Considerations	11
4.8	Design Limitations	11
5	Manufacturing and Material Considerations	12
5.1	Material Selection Rationale	12
5.2	Manufacturing Approach	12
5.3	Structural Assembly Considerations	12
5.4	Design for Manufacturability	13
5.5	Material and Manufacturing Limitations	13

6	Limitations and Future Work	13
6.1	Design Scope Limitations	13
6.2	Opportunities for Further Development	13
6.3	Portfolio and Professional Context	14
7	Landing Gear Conceptual Design	14
7.1	Design Objectives	14
7.2	Conceptual Geometry	14
7.3	Structural Load Path	15
7.4	Section Analysis	16
7.5	Orthographic Representation	16
7.6	Design Scope and Limitations	17
7.7	Summary	17
8	Conclusion	17

1 Introduction

The wing is one of the most highly loaded structural components of a fixed-wing aircraft. During flight, it must resist significant bending moments generated by aerodynamic lift, as well as shear and torsional loads arising from maneuvering, gusts, and asymmetric conditions. The primary structure responsible for carrying these loads is the wing box.

The wing box typically consists of upper and lower load-carrying skins connected by spars and internal structure, forming a closed-section beam. This configuration provides high bending stiffness and torsional rigidity while maintaining structural efficiency. As a result, the wing box plays a central role in determining the overall strength, stiffness, and weight of the wing.

The objective of this project is to develop a conceptual structural design of an aircraft wing box suitable for early-stage aerospace design evaluation. Rather than focusing on detailed stress sizing or certification-level analysis, this study emphasizes structural layout, load-path reasoning, and design justification. A parametric CAD model is used to support the structural concept and to visually demonstrate how loads are transmitted through the structure.

This project is intended as a portfolio-quality engineering study, demonstrating the ability to reason about aircraft primary structures, make defensible design assumptions, and communicate engineering decisions clearly. The scope reflects realistic industry practice at the conceptual design level, where geometry, architecture, and structural philosophy are established prior to detailed numerical analysis.

The report is organized as follows. Section 2 defines the design context and key assumptions used throughout the study. Section 3 presents representative flight load cases used to inform the structural design. Section 4 details the wing box geometry and internal structure, supported by CAD visualizations. Manufacturing considerations, limitations, and opportunities for future refinement are discussed in subsequent sections before concluding remarks are provided.

2 Design Context and Assumptions

2.1 Design Context

In order to develop a meaningful structural design, it is necessary to define a reference context that reflects realistic aircraft operating conditions without constraining the study to a specific certified platform. For this project, the wing box is designed within the context of a conventional fixed-wing aircraft operating in the subsonic flight regime.

The intent is not to represent a finalized aircraft configuration, but rather to establish a credible structural environment in which primary wing loads are generated and trans-

mitted. This approach is consistent with early-stage aerospace design practice, where structural architecture is explored before detailed performance sizing is undertaken.

The wing is assumed to be the primary lift-generating surface and therefore the dominant structural driver. As such, the wing box is treated as the main load-carrying component responsible for resisting bending, shear, and torsional loads.

2.2 Reference Aircraft Characteristics

To support the conceptual design, the reference aircraft is assumed to exhibit the following general characteristics:

- Conventional fixed-wing configuration
- Cantilever wing attached to the fuselage
- Subsonic operating envelope
- Symmetric loading under normal flight conditions

These characteristics are representative of a broad class of transport and utility aircraft and provide a reasonable basis for preliminary structural design decisions.

2.3 Structural Design Philosophy

The structural design philosophy adopted in this study prioritizes clarity of load paths, structural efficiency, and manufacturability. The wing box is treated as a closed-section beam, with the upper and lower skins acting as primary bending members and the spar webs carrying shear loads and maintaining geometric separation between the skins.

Rather than optimizing individual structural elements, the focus is placed on establishing a realistic and defensible overall geometry. This philosophy aligns with conceptual and preliminary design stages, where architectural decisions have a greater impact on structural performance than local sizing refinements.

2.4 Material Assumptions

For the purposes of this conceptual design, the wing box structure is assumed to be constructed from a conventional aerospace-grade aluminum alloy. Aluminum alloys remain widely used in aircraft primary structures due to their favorable strength-to-weight ratio, established manufacturing methods, and predictable failure behavior.

Material behavior is assumed to be linear elastic, and effects such as plastic deformation, fatigue, and environmental degradation are not explicitly considered. These assumptions are appropriate for an early-stage structural layout study.

2.5 Geometric and Modeling Assumptions

The wing box is modeled as a straight, prismatic section representing a portion of the wing away from complex geometric features such as taper, sweep variation, and root fittings. Internal ribs, fasteners, and local reinforcements are not explicitly modeled.

This simplified representation allows the primary load-carrying behavior of the wing box to be examined without introducing unnecessary complexity. Detailed geometric features are more appropriately addressed during later stages of structural design and analysis.

2.6 Scope and Limitations

The assumptions presented in this section define the scope of the study and establish clear boundaries for the conclusions that can be drawn. The design is intended to demonstrate structural reasoning and architectural decision-making rather than provide certification-ready sizing.

Future work would be required to refine the design through detailed loads analysis, stress sizing, and validation against regulatory requirements.

3 Flight Loads and Design Cases

3.1 Purpose of Load Definition

Structural design decisions must be informed by representative loading conditions. For an aircraft wing, the dominant loads arise from aerodynamic lift, which generates bending moments, shear forces, and torsional effects along the span. The objective of this section is to define a set of conceptual load cases that establish the structural demands placed on the wing box.

Rather than performing a detailed loads analysis, this study identifies representative maneuver and flight conditions that are commonly used to guide early-stage wing structural design. These load cases are subsequently used to justify the wing box geometry and internal load paths presented in Section 4.

3.2 Load Factor Concept

The load factor, denoted by n , is defined as the ratio of aerodynamic lift to aircraft weight:

$$n = \frac{L}{W} \tag{1}$$

where L represents the lift force generated by the wing and W is the aircraft weight. A load factor greater than unity indicates that the wing is supporting loads greater than those experienced during steady level flight.

Load factor provides a convenient means of expressing maneuver-induced structural demand and is widely used in conceptual and preliminary aircraft structural design.

3.3 Representative Design Load Factors

To establish credible structural design conditions, representative limit load factors were assumed based on typical values for conventional subsonic aircraft. The following values were selected for conceptual design purposes:

- Positive limit load factor: $n_+ = +3.8$
- Negative limit load factor: $n_- = -1.5$

These values are consistent with commonly referenced certification standards for normal category aircraft and provide a reasonable basis for preliminary wing structural design.

3.4 V–n Diagram Interpretation

The velocity–load factor (V–n) diagram provides a graphical representation of the aircraft operating envelope by relating allowable load factor to airspeed. At low airspeeds, the achievable load factor is limited by aerodynamic stall, while at higher airspeeds the structure becomes the limiting factor.

For the purposes of this study, the V–n diagram is used conceptually to identify critical structural design cases rather than to define exact operating boundaries. The intersection of maximum load factor and high airspeed represents the most demanding bending condition for the wing structure.

3.5 Critical Structural Design Cases

Based on the conceptual V–n envelope, the following load cases were identified as structurally significant:

- Positive maneuvering load at high airspeed ($+3.8g$)
- Negative maneuvering load resulting in wing down-bending ($-1.5g$)
- Symmetric lift-induced bending during steady level flight

Among these cases, the positive maneuvering load produces the largest bending moment at the wing root and is therefore the primary driver for wing box design.

3.6 Implications for Wing Box Design

Under positive load factor conditions, the wing experiences upward aerodynamic lift, resulting in a bending moment that places the upper wing surface in compression and the lower surface in tension. The wing box must therefore provide sufficient separation between the upper and lower skins to efficiently resist bending stresses.

Shear loads generated by lift are transferred between the skins through the spar webs, while the closed-section geometry provides torsional rigidity and resistance to twist. These structural requirements directly inform the wing box geometry and internal structure developed in the following section.

3.7 Limitations of Load Definition

The load cases presented in this section are intended for conceptual design guidance only. Gust loading, asymmetric lift, and dynamic effects are not explicitly considered. A complete structural certification effort would require a comprehensive loads analysis in accordance with applicable regulatory standards.

Despite these limitations, the selected load cases provide a realistic and defensible basis for the preliminary wing box design presented in this study.

4 Wing Box Structural Design

4.1 Role of the Wing Box

The wing box serves as the primary load-carrying structural element of the aircraft wing. It is responsible for resisting bending moments generated by aerodynamic lift, transferring shear loads between structural members, and providing torsional rigidity to prevent excessive wing twist.

In conceptual aircraft design, the wing box architecture has a dominant influence on structural performance. Decisions regarding spar placement, skin separation, and cross-sectional geometry largely determine bending stiffness, torsional stiffness, and overall structural efficiency. As such, the wing box is treated as the central structural focus of this study.

4.2 Structural Concept

The wing box was designed as a closed-section beam consisting of upper and lower skins connected by front and rear spar webs. This configuration is widely used in aircraft primary structures due to its ability to efficiently resist combined bending and torsional loads.

A hollow interior was incorporated to reduce structural weight while maintaining stiffness. The resulting geometry provides a favorable balance between load-carrying capability and material efficiency, consistent with early-stage aerospace structural design practice.

4.3 Geometric Configuration

The wing box geometry was developed using parametric CAD modeling. A straight, prismatic section was selected to represent a portion of the wing away from geometric discontinuities such as the wing root and control surface cutouts.

Figure 1 presents an isometric view of the wing box, illustrating the overall configuration and closed-section geometry.

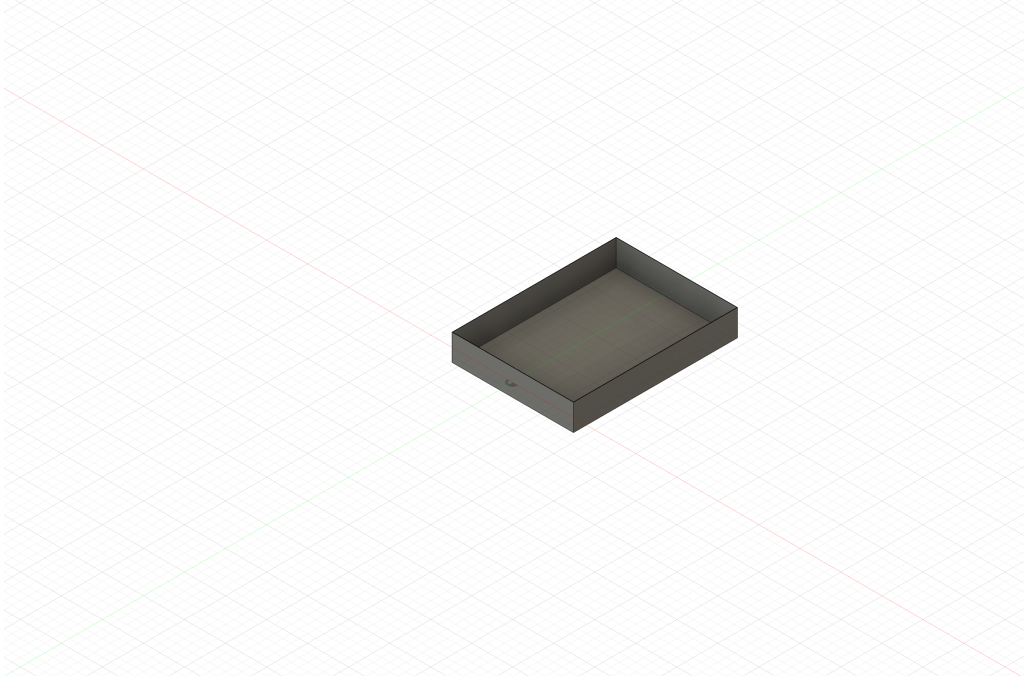


Figure 1: Isometric view of the wing box structure

4.4 Planform and Cross-Section Views

The planform and front views of the wing box are shown in Figures 2 and 3, respectively. These views highlight the spanwise extent of the structure and the separation between the spar webs.

The depth of the wing box plays a critical role in resisting bending moments, as increased separation between the upper and lower skins leads to reduced bending stress for a given load. Spar placement is therefore a key design variable influencing structural performance.

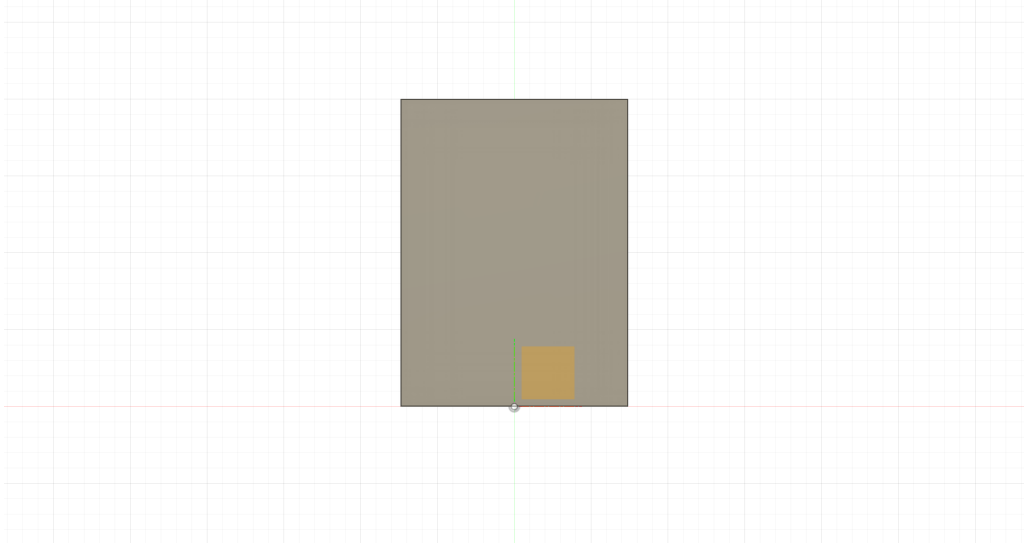


Figure 2: Top view of the wing box geometry

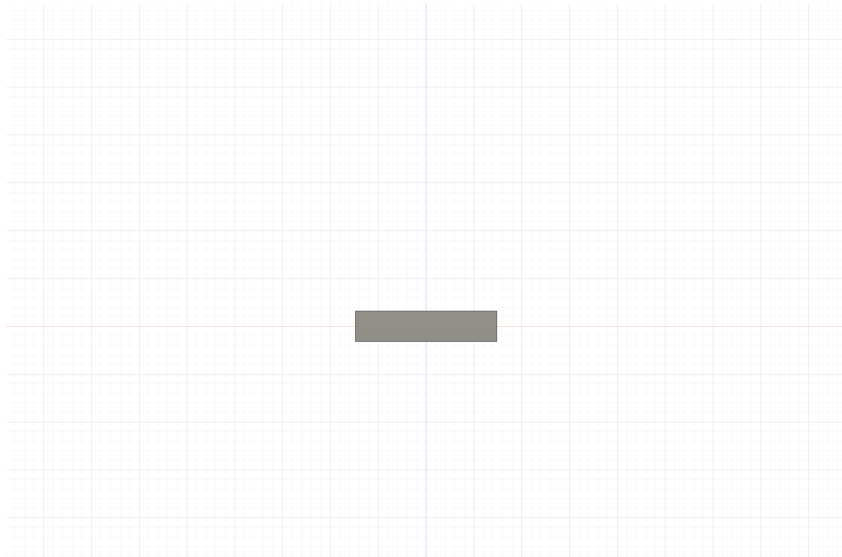


Figure 3: Front view of the wing box showing spar spacing and box depth

4.5 Internal Structure and Section Analysis

A section analysis was performed within the CAD environment to visualize the internal structure of the wing box. The resulting cross-sectional view is shown in Figure 4.

This view clearly illustrates the hollow interior of the wing box and the load-carrying skins and spar webs. The closed-section geometry provides significant torsional rigidity, which is essential for maintaining aerodynamic performance and preventing excessive wing twist under load.

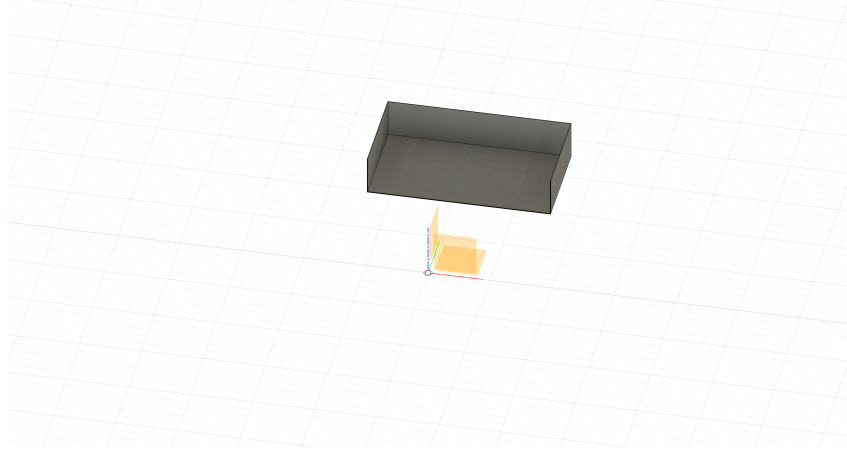


Figure 4: Section view of the wing box highlighting internal load-carrying structure

4.6 Load Path Discussion

Under positive maneuvering loads, aerodynamic lift induces an upward bending moment along the wing span. This bending places the upper skin of the wing box in compression and the lower skin in tension. The spar webs primarily carry shear loads and maintain the geometric separation between the skins.

The closed-section nature of the wing box also enables efficient resistance to torsional loads. Torque generated by aerodynamic moments is distributed around the perimeter of the section, reducing localized stress concentrations and improving overall structural stability.

4.7 Structural Efficiency Considerations

The selected wing box configuration prioritizes structural efficiency by concentrating material in regions that contribute most effectively to load carrying. By maximizing skin separation and maintaining a hollow interior, the structure achieves high bending stiffness with minimal material usage.

This approach reflects standard aerospace structural design philosophy, where weight efficiency is achieved through geometry and load-path optimization rather than excessive material thickness.

4.8 Design Limitations

The wing box design presented in this section is intended for conceptual demonstration purposes. Detailed stress sizing, buckling analysis, and fatigue assessment were not performed. Additionally, internal ribs, fasteners, and local reinforcements were not explicitly modeled.

These aspects represent important areas for future refinement but are beyond the scope of the present conceptual design study.

5 Manufacturing and Material Considerations

5.1 Material Selection Rationale

For the purposes of this conceptual structural design study, the wing box is assumed to be constructed from a conventional aerospace-grade aluminum alloy. Aluminum alloys are widely used in aircraft primary structures due to their favorable strength-to-weight ratio, predictable elastic behavior, corrosion resistance, and well-established manufacturing processes.

While advanced composite materials are increasingly used in modern aircraft, aluminum remains an appropriate and widely accepted baseline material for conceptual design studies. The use of aluminum also simplifies manufacturing assumptions and facilitates clearer structural load-path reasoning.

Material properties are assumed to be linear elastic, and material selection is treated at a high level consistent with early-stage design practice.

5.2 Manufacturing Approach

The wing box geometry developed in this study is compatible with conventional aerospace manufacturing methods. The structure may be fabricated using a combination of sheet metal skins and machined or formed spar webs.

The upper and lower skins would typically be manufactured as formed aluminum panels, while the spar webs could be produced through machining or extrusion, depending on design requirements. Assembly would be achieved using mechanical fasteners or bonded joints, consistent with standard aircraft production practices.

The simplified, prismatic geometry of the wing box facilitates ease of fabrication, inspection, and structural integration.

5.3 Structural Assembly Considerations

The wing box configuration allows for a modular assembly process in which individual structural elements are manufactured separately and assembled into a closed-section structure. This approach supports:

- Efficient load transfer between skins and spars
- Ease of inspection and maintenance

- Flexibility for future structural modifications

Although ribs and attachment fittings are not explicitly modeled in this study, the wing box geometry readily accommodates their inclusion in later design stages.

5.4 Design for Manufacturability

From a manufacturability perspective, the wing box design prioritizes simplicity and repeatability. Straight spar webs, uniform skin thickness, and a constant cross-section reduce manufacturing complexity and minimize production risk.

These characteristics are particularly advantageous during early development phases, where design flexibility and rapid iteration are valued. The geometry can be readily adapted to include taper, sweep, or local reinforcements as the design matures.

5.5 Material and Manufacturing Limitations

The material and manufacturing considerations presented in this section are intentionally high-level. Detailed material selection, fastener design, tolerance analysis, and production cost modeling were not performed.

These elements represent critical aspects of later-stage structural development and would be addressed during detailed design and certification phases.

6 Limitations and Future Work

6.1 Design Scope Limitations

The wing box design presented in this report is intended as a conceptual structural study. As such, several aspects of detailed aircraft structural design were intentionally excluded to maintain focus on structural layout and load-path reasoning.

No finite element stress analysis was performed, and skin thicknesses and spar dimensions were not optimized through numerical sizing. Buckling behavior, fatigue life, and damage tolerance were also not evaluated. Additionally, internal ribs, fasteners, and attachment fittings were not explicitly modeled in the CAD geometry.

These limitations are consistent with early-stage design practice, where architectural decisions are established prior to detailed structural sizing and certification analysis.

6.2 Opportunities for Further Development

The wing box architecture developed in this study provides a strong foundation for future refinement. Potential extensions of this work include:

- Finite element analysis to evaluate stress distribution and identify critical load paths
- Buckling analysis of compression skins under positive maneuvering loads
- Material trade studies comparing aluminum and composite constructions
- Integration of internal ribs and wing root attachment features
- Weight optimization and structural efficiency assessment

These developments would enable progression from conceptual design toward a preliminary or detailed structural design phase.

6.3 Portfolio and Professional Context

This project is intended to serve as a portfolio-quality demonstration of structural reasoning, CAD execution, and engineering communication. The modular structure of the design and documentation allows for incremental upgrades as additional analysis capabilities are developed.

As such, the project reflects a realistic engineering workflow rather than a finalized or certification-ready design.

7 Landing Gear Conceptual Design

7.1 Design Objectives

The landing gear system is a critical structural subsystem responsible for supporting the aircraft during ground operations, absorbing landing loads, and transferring ground reaction forces into the primary structure. The objective of this design is to present a conceptual landing gear configuration that demonstrates realistic geometry, clear load paths, and appropriate engineering judgment suitable for an early-stage aircraft design.

This design focuses on structural representation rather than detailed mechanisms, with emphasis placed on clarity, proportional realism, and effective communication of load transfer.

7.2 Conceptual Geometry

The landing gear configuration consists of a cylindrical wheel and housing assembly connected to a rectangular strut element. Two configurations are modeled: a primary main landing gear and a smaller nose landing gear. The geometry is intentionally simplified to represent the dominant structural features without unnecessary detail.

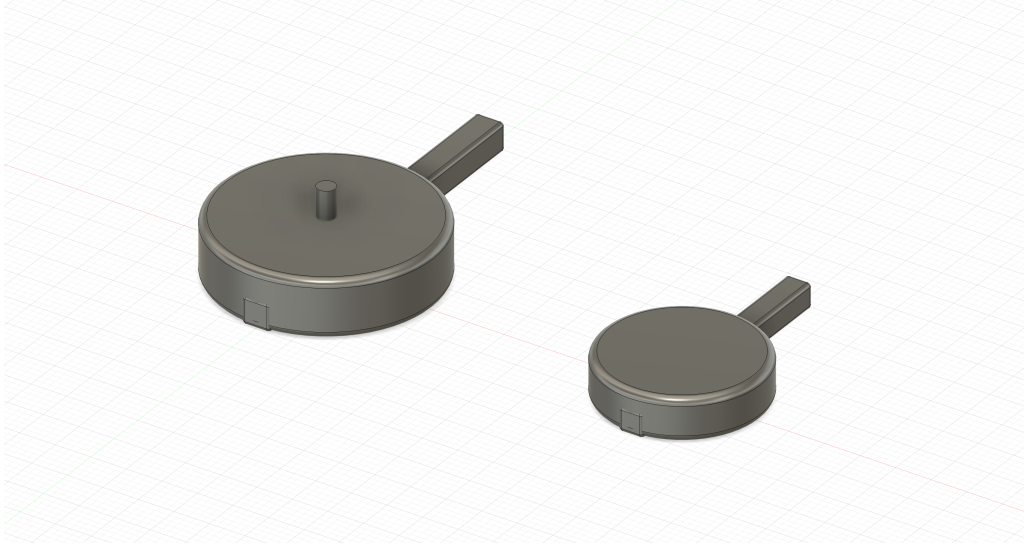


Figure 5: Isometric view of the conceptual main and nose landing gear assemblies.

7.3 Structural Load Path

During landing and ground operations, vertical ground reaction forces act at the wheel-ground interface and are transmitted through the wheel housing into the strut. The strut carries these loads primarily in compression and bending, transferring forces into the aircraft structure.

The rectangular strut geometry provides increased bending stiffness compared to a circular cross-section of similar mass, making it suitable for resisting landing-induced moments.

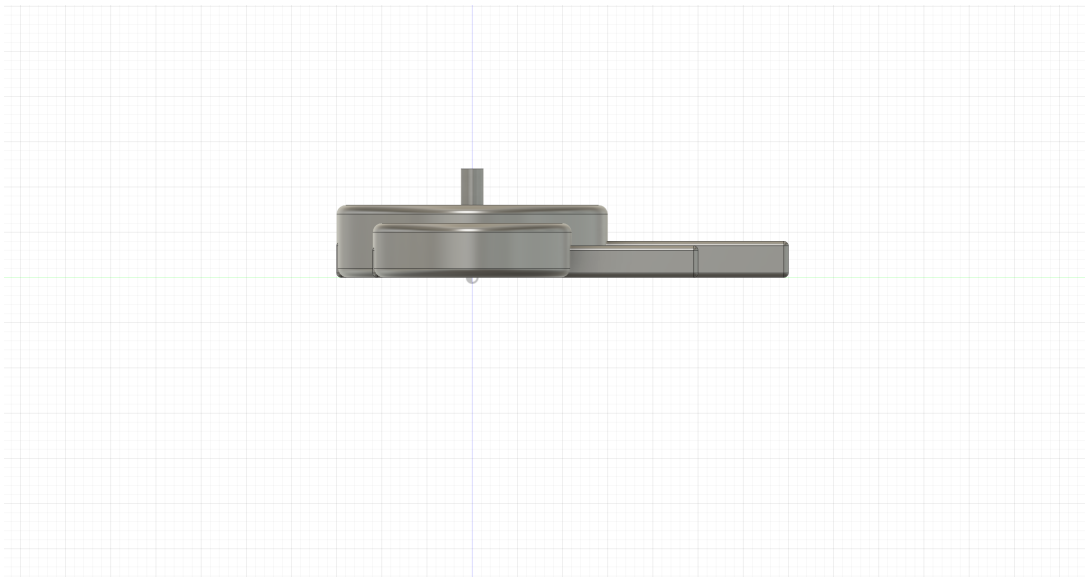


Figure 6: Side view of the landing gear assembly illustrating the primary load direction and strut orientation.

7.4 Section Analysis

A section analysis was performed through the landing gear assembly to visualize internal geometry and load transfer paths. The sectioned view highlights material continuity through the wheel housing and strut, providing insight into how landing loads are distributed through the component volume.

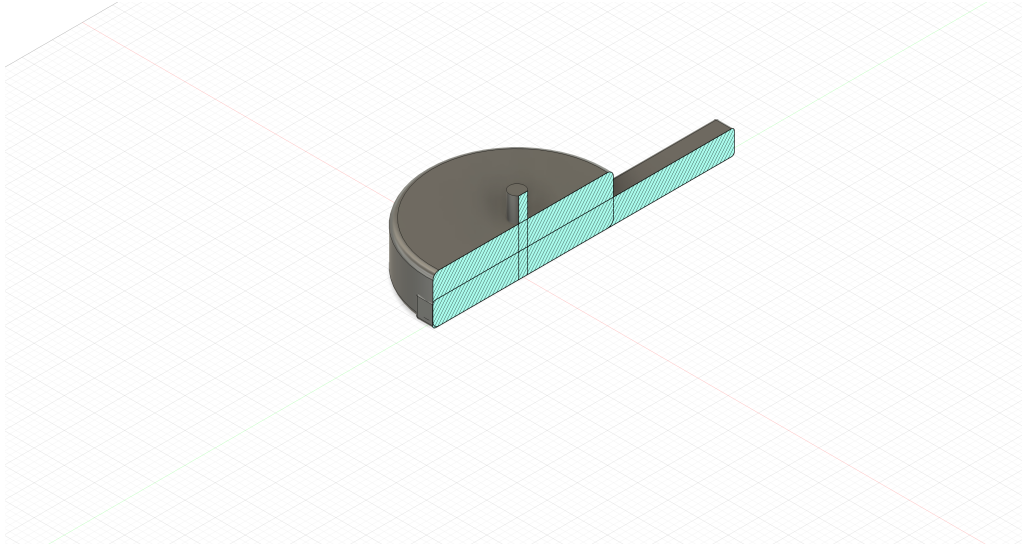


Figure 7: Section view of the landing gear assembly showing internal structure and load transfer.

7.5 Orthographic Representation

Orthographic views are included to support clear geometric interpretation and technical communication. These views remove perspective distortion and allow the landing gear geometry to be evaluated in a standard engineering format.

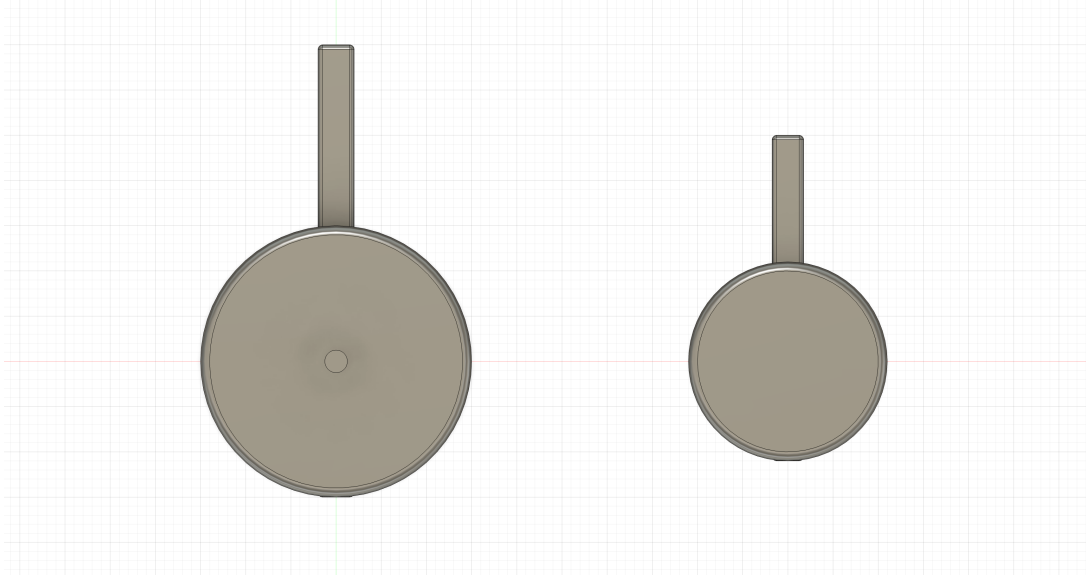


Figure 8: Front view of the landing gear assembly highlighting symmetry and wheel geometry.

7.6 Design Scope and Limitations

The landing gear design presented is limited to conceptual structural representation. Detailed shock absorption mechanisms, fatigue life analysis, steering systems, and attachment fittings are outside the scope of this stage and would be addressed during subsequent detailed design phases.

7.7 Summary

The conceptual landing gear design demonstrates a structurally coherent and proportionally realistic solution suitable for early-stage aircraft design. The use of simplified geometry, section analysis, and orthographic views effectively communicates structural intent and load paths while maintaining appropriate modeling restraint.

8 Conclusion

A conceptual structural design study of an aircraft wing box has been presented. The work focused on establishing a realistic wing box architecture capable of resisting representative bending, shear, and torsional loads encountered during flight.

By defining a credible design context and representative load cases, a closed-section wing box geometry was developed that reflects conventional aerospace structural practice. The use of parametric CAD modeling enabled clear visualization of internal load paths and supported the structural reasoning presented throughout the report.

While detailed stress sizing and certification-level analysis were beyond the scope of this study, the resulting design demonstrates a sound understanding of aircraft primary structures and early-stage structural decision-making. The project provides a strong foundation for future refinement through numerical analysis, material optimization, and integration of additional structural features.

Overall, this work serves as a portfolio-quality demonstration of aerospace structural design principles, engineering judgment, and technical communication.